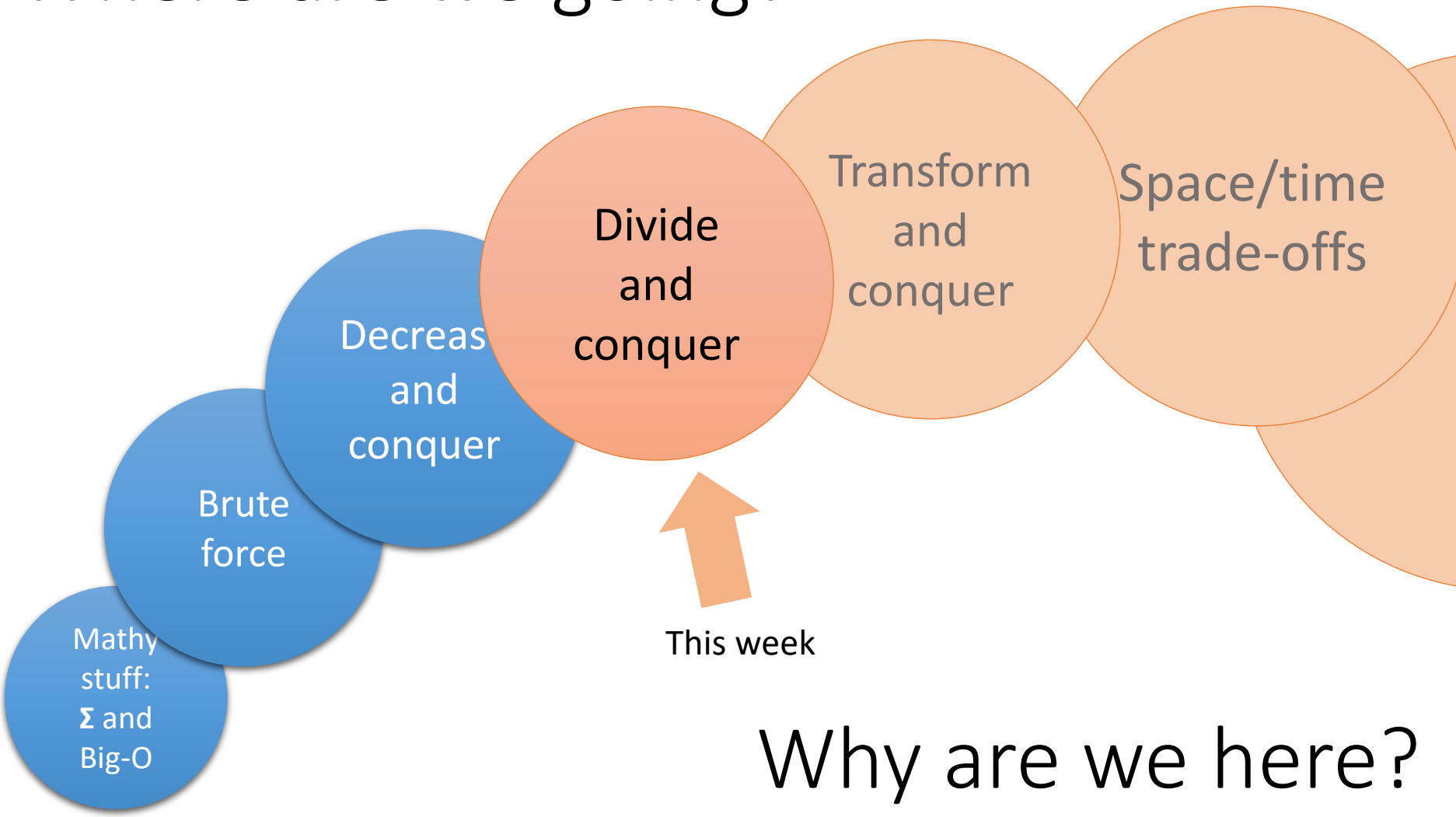
Where have we been?
Where are we going?



Why are we here?

No. Too deep.

This week:

- *Divide and conquer* algorithms
- Example: Count a key in an array
- How to analyze Divide and Conquer (the “Master Theorem”)
- Example: Mergesort

First a bit about recursion

Our old friend Factorial()

```
Factorial(n)
 if n=0 or n=1 then
   return 1
 else
   return n * Factorial(n - 1)
```

```
Factorial(5)
 if n=0 or n=1 then
   return 1
 else
   return 5 *
```

```
Factorial(4)
 if n=0 or n=1 then
   return 1
 else
   return
     4 *
```

```
Factorial(3)
 if n=0 or n=1 then
   return 1
 else
   return
     3 *
```

```
Factorial(2)
 if n=0 or n=1 then
   return 1
 else
   return
     2 *
```

```
Factorial(1)
 if n=0 or n=1 then
   return 1
 else
   return 1 * Factorial(0)
```

```
Factorial(5)
 if n=0 or n=1 then
   return 1
 else
   return 5 *
```

```
Factorial(4)
 if n=0 or n=1 then
   return 1
 else
   return
     4 *
```

```
Factorial(3)
 if n=0 or n=1 then
   return 1
 else
   return
     3 *
```

```
Factorial(2)
 if n=0 or n=1 then
   return 1
 else
   return
     2 *
```

```
Factorial(1)
 if n=0 or n=1 then
   return 1
```

```
Factorial(5)
 if n=0 or n=1 then
   return 1
 else
   return 5 *
```

```
Factorial(4)
 if n=0 or n=1 then
   return 1
 else
   return
     4 *
```

```
Factorial(3)
 if n=0 or n=1 then
   return 1
 else
   return
     3 *
```

```
Factorial(2)
 if n=0 or n=1 then
   return 1
 else
   return
     2 * 1
```

```
Factorial(5)
 if n=0 or n=1 then
   return 1
 else
   return 5 *
```

```
Factorial(4)
 if n=0 or n=1 then
   return 1
 else
   return
     4 *
```

```
Factorial(3)
 if n=0 or n=1 then
   return 1
 else
   return
     3 *
```

```
Factorial(2)
 if n=0 or n=1 then
   return 1
 else
   return 2
```

```
Factorial(5)
 if n=0 or n=1 then
   return 1
 else
   return 5 *
```

```
Factorial(4)
 if n=0 or n=1 then
   return 1
 else
   return
     4 *
```

```
Factorial(3)
 if n=0 or n=1 then
   return 1
 else
   return
     3 * 2
```

```
Factorial(5)
 if n=0 or n=1 then
   return 1
 else
   return 5 *
```

```
Factorial(4)
 if n=0 or n=1 then
   return 1
 else
   return
     4 *
```

```
Factorial(3)
 if n=0 or n=1 then
   return 1
 else
   return 6
```

```
Factorial(5)
 if n=0 or n=1 then
   return 1
 else
   return 5 *
```

```
Factorial(4)
 if n=0 or n=1 then
   return 1
 else
   return
     4 * 6
```

```
Factorial(5)
 if n=0 or n=1 then
   return 1
 else
   return 5 *
```

```
Factorial(4)
 if n=0 or n=1 then
   return 1
 else
   return 24
```

```
Factorial(5)
 if n=0 or n=1 then
   return 1
 else
   return 5 * 24
```

```
Factorial(5)
 if n=0 or n=1 then
   return 1
 else
   return 120
```

```
Factorial(5) ==> 120
```

Example: permutations

```
generatePerms (a1, a2, ..., an)
   if n > 1
      smallerPerms = generatePerms (a1, a2, ..., an-1)
      initialize allPerms to {}
      for each p in smallerPerms
         insert an before a1 and add to allPerms
         for i = 1 to n-1
            insert an after ai and add to allPerms
      return allPerms
```

Permutations algorithm

```
generatePerms (a1, a2, ..., an)
   if n > 1
      smallerPerms =
              generatePerms (...)

                 /* it does whatever it does */


      initialize allPerms to {}
      for each p in smallerPerms
         insert an before a1 and add to allPerms
         for i = 1 to n-1
            insert an after ai and add to allPerms
      return allPerms
```

Permutations algorithm

```
generatePerms (a1, a2, ..., an)
   if n > 1
      smallerPerms =

              {perm, perm, perm, perm, ..., perm}




      initialize allPerms to {}
      for each p in smallerPerms
         insert an before a1 and add to allPerms
         for i = 1 to n-1
            insert an after ai and add to allPerms
      return allPerms
```

Lecture 4

COMP 3760

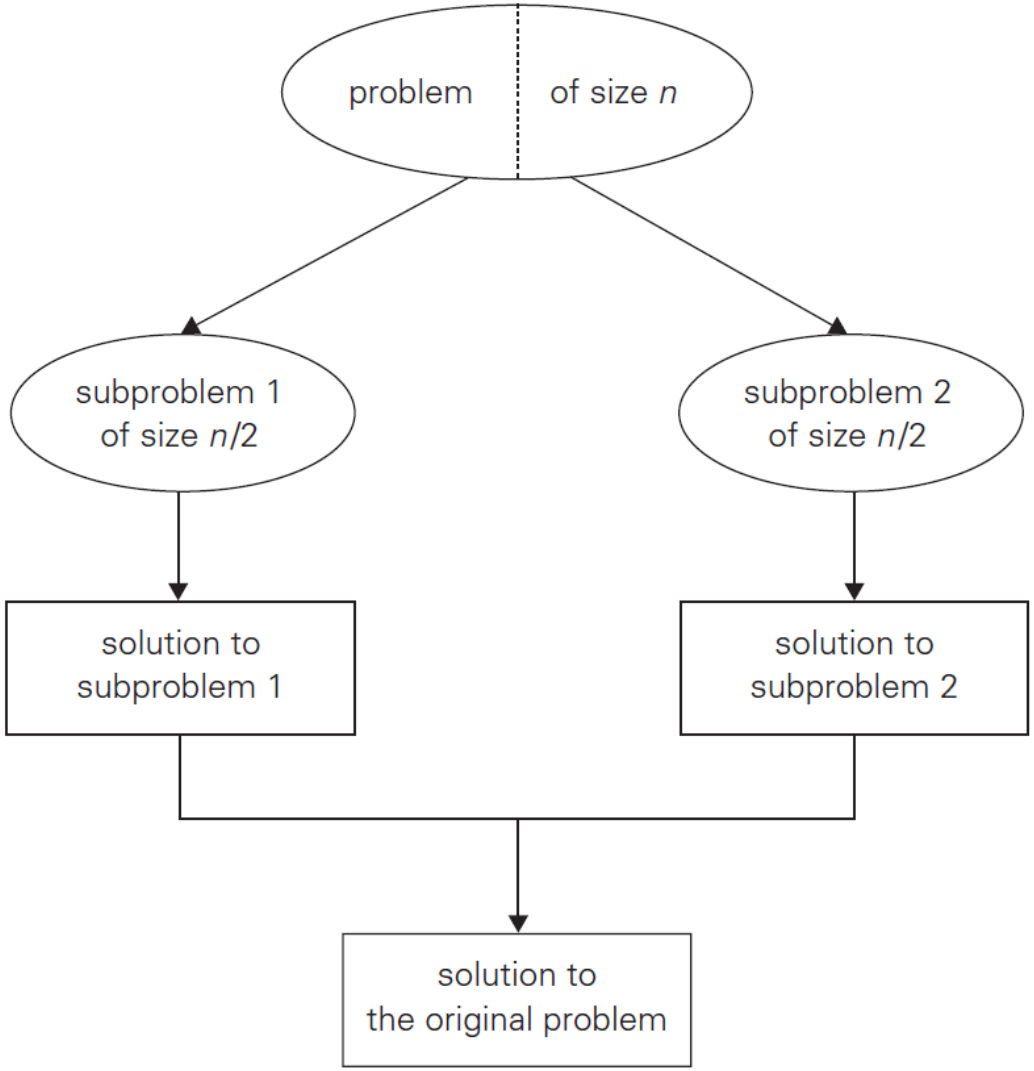
Divide and Conquer algorithms

Text sections 5.1, 5.3

Divide and conquer algorithms

- Divide a problem into two or more smaller instances
- Solve smaller instances (often recursively)
- Obtain solution to original (larger) instance by combining these solutions

Divide and conquer technique



Divide-and-conquer vs. decrease-and-conquer

- Think of the fake coin problem (decrease-and-conquer):
  - We discarded half the coins at each step
  - So we only worked on one "subproblem"

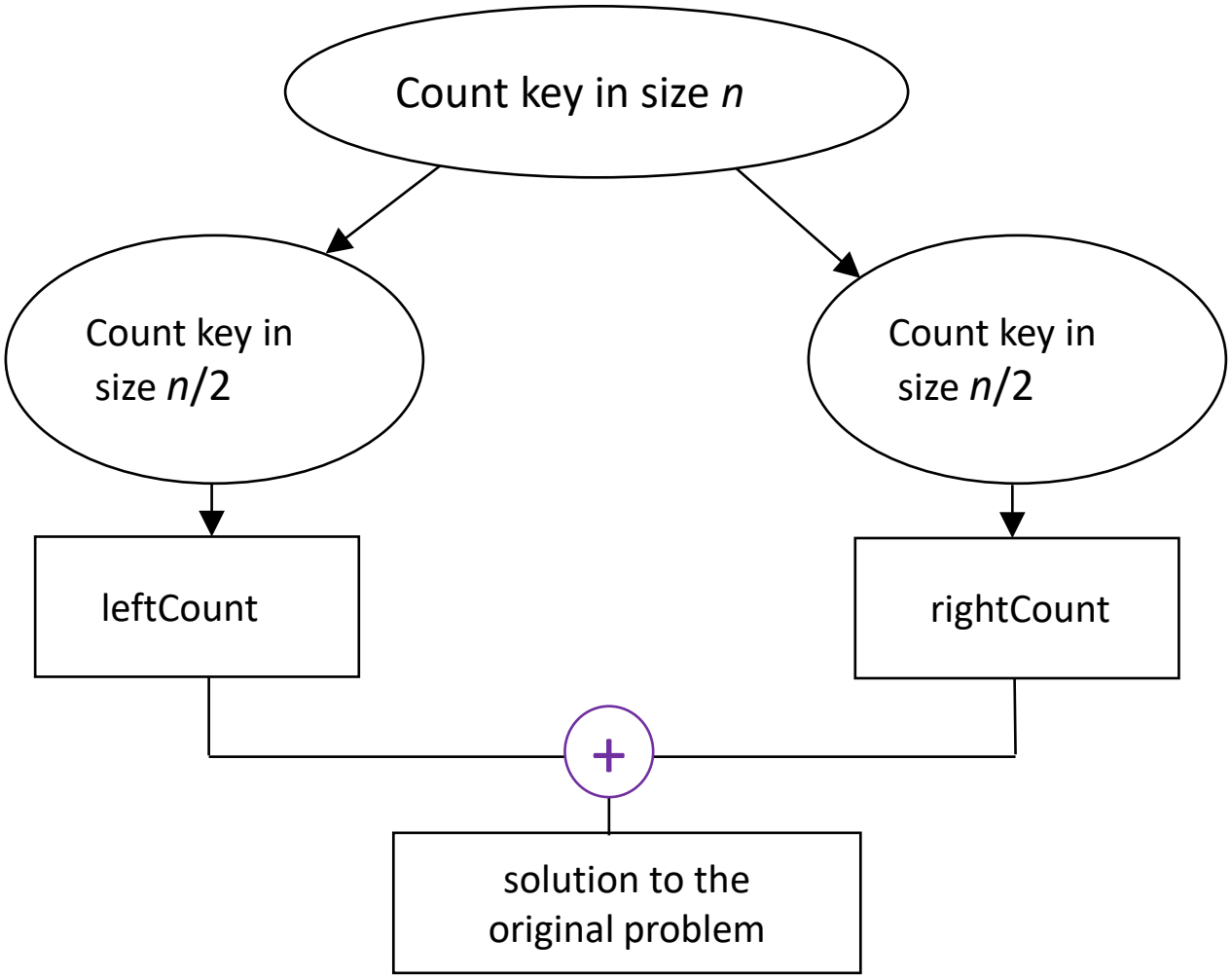
- For divide and conquer…
  - You will solve <u>multiple</u> subproblems

Example:
Count a key in an array

Count a key in an array

- Problem:
  - Count the number of times a specific key occurs in an array.
- For example:
  - If input array is A=[2,7,6,6,2,4,6,9,2] and key=6…
  - … should return the value 3.
- Design an algorithm that uses divide and conquer

Count a key in an array



Count a key in an array

```
Algorithm CountKeys(A[], Key, L, R)

//Input: A[] is an array A[0..n-1]
//      L & R (L ≤ R) are boundaries of the current search
//Output: The number of times Key exists in A[L..R]

if L = R
   if (A[L] = Key) return 1
   else return 0
else
   leftCount = CountKey(A[], Key,  L,     ⌊(L+R)/2⌋)
   rightCount = CountKey(A[], Key,  ⌊(L+R)/2⌋+1,  R)
   return leftCount + rightCount
```

Count a key in an array

- Superficially, CountKeys resembles Binary Search
  - Similar arguments (array bounds)
  - Finding a midpoint
  - What's the difference?

- We have to process both sides
  - In CountKeys, both sides must be searched
  - In Binary Search, one half gets ignored

Analysis of divide and conquer

Analyzing a divide-and-conquer algorithm

- What matters:
  1. Number of parts — a
  2. Size of each part — n/b
  3. Cost of "extra work" — $F(n)$

This expression is your new friend:

$$n^{\log_b a}$$

Count a key in an array

```
Algorithm CountKeys(A[], Key, L, R)

//Input: A[] is an array A[0..n-1]
//       L & R (L ≤ R) are boundaries of the current search
//Output: The number of times Key exists in A[L..R]

if L = R
    if (A[L] = Key) return 1
    else return 0
else
    leftCount = CountKey(A[], Key,  L,      ⌊(L+R)/2⌋)
    rightCount = CountKey(A[], Key,  ⌊(L+R)/2⌋+1,  R)
    return leftCount + rightCount
```

2 subproblems

i.e., 2 recursive calls

a = 2

Count a key in an array

```
Algorithm CountKeys(A[], Key, L, R)

//Input: A[] is an array A[0..n-1]
//       L & R (L ≤ R) are boundaries of the current search
//Output: The number of times Key exists in A[L..R]

if L = R
    if (A[L] = Key) return 1
    else return 0
else
    leftCount = CountKey(A[], Key, L,      ⌊(L+R)/2⌋)
    rightCount = CountKey(A[], Key, ⌊(L+R)/2⌋+1,  R)
    return leftCount + rightCount
```

each subproblem is half the size (n/2)

b = 2

Count a key in an array

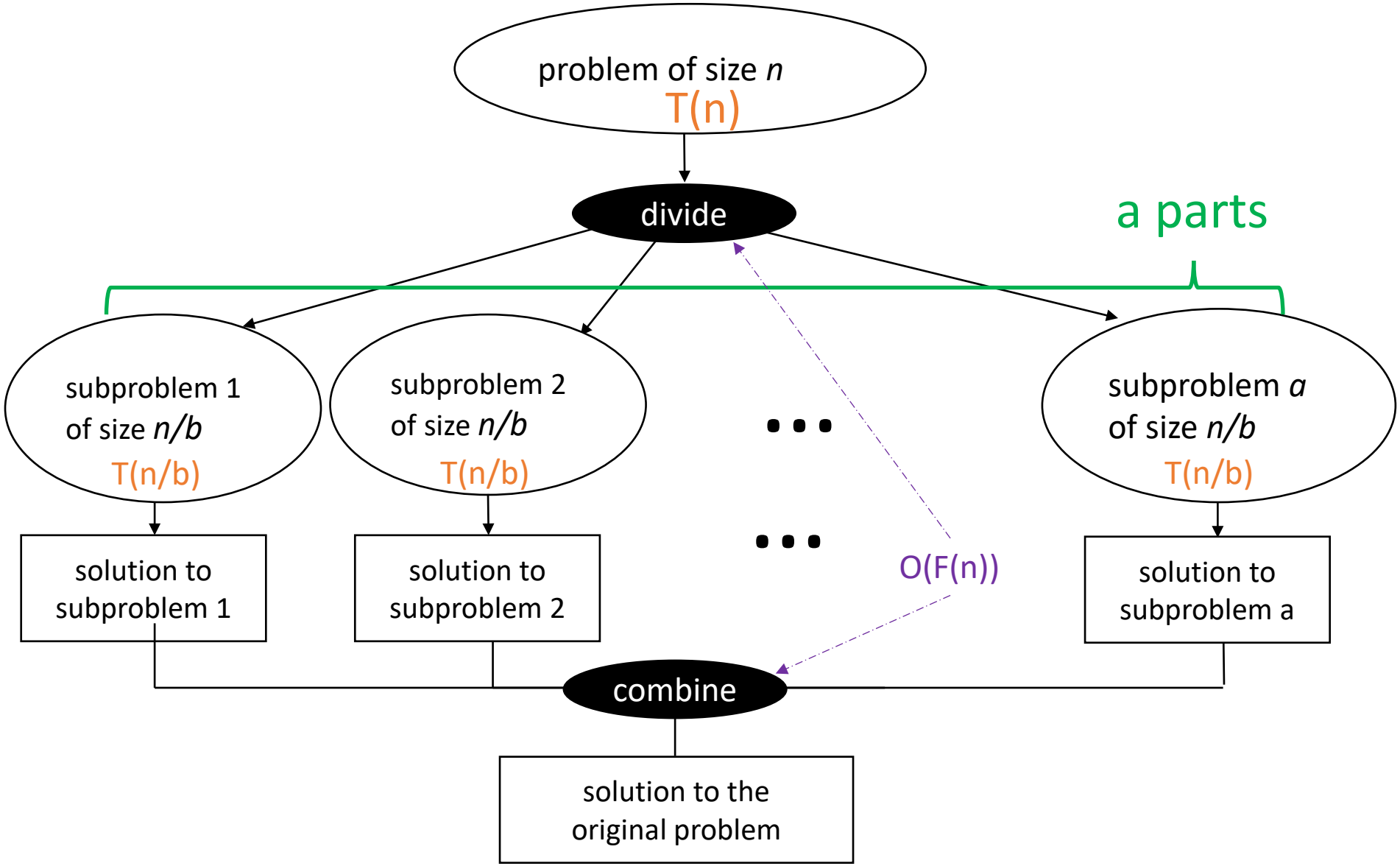
```
Algorithm CountKeys(A[], Key, L, R)

//Input: A[] is an array A[0..n-1]
//      L & R (L ≤ R) are boundaries of the current search
//Output: The number of times Key exists in A[L..R]

if L = R
    if (A[L] = Key) return 1
    else return 0
else
    leftCount = CountKey(A[], Key,  L,     ⌊(L+R)/2⌋)
    rightCount = CountKey(A[], Key,  ⌊(L+R)/2⌋+1,  R)
    return leftCount + rightCount
```

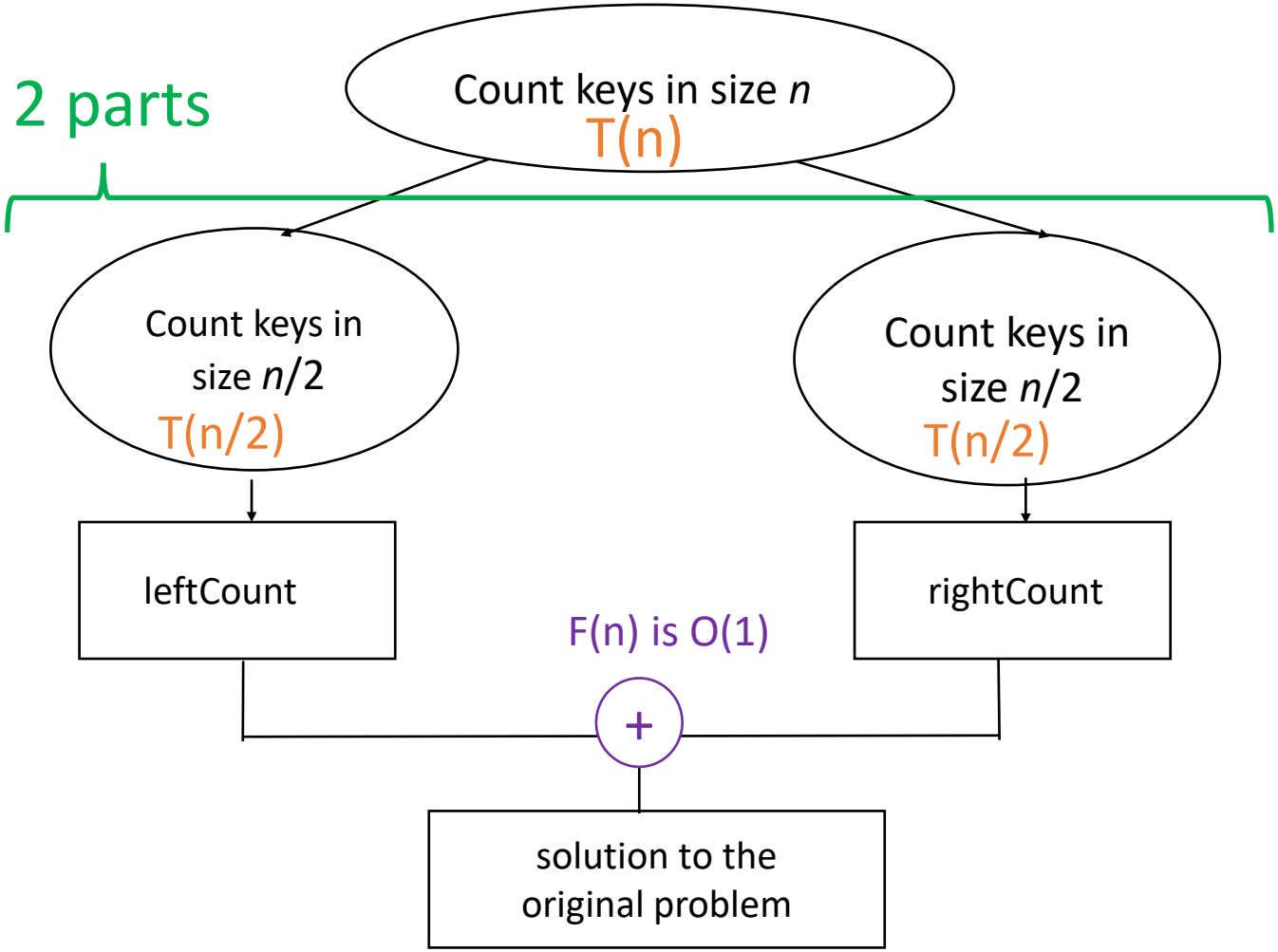
additional computation
time is O(1)
F(n) = 1

Analysis of a divide and conquer algorithm



$$T(n) = a\,T(n/b) + F(n)$$

Example: analysis of CountKeys



$$T(n) = 2\,T(n/2) + 1$$

O(*???*)

What is the big-O efficiency class of T(n)?

$T(n) = a\,T(n/b) + F(n)$

1 Compare $n^{\log_b a}$ and $F(n)$

2a The bigger one wins

2b If they're equal: $O(n^{\log_b a} \log n)$

The Master Theorem

If $T(n) = a\,T(n/b) + F(n)$

1) If $n^{\log_b a} < F(n)$, $T(n) \in O(F(n))$
2) If $n^{\log_b a} > F(n)$, $T(n) \in O(n^{\log_b a})$
3) If $n^{\log_b a} = F(n)$, $T(n) \in O(n^{\log_b a} \log n)$

Master Theorem examples

Example 1: $T(n) = 4T(n/2) + n \Rightarrow T(n) \in ?$

a = 4
b = 2
F(n) = n

$\rightarrow n^{\log_b a} \rightarrow n^{\log_2 4} \rightarrow n^2$

F(n) = n

$\rightarrow T(n) \in O(n^2)$

Example 2: $T(n) = 4T(n/2) + n^2 \Rightarrow T(n) \in ?$

a = 4
b = 2
F(n) = n^2

$\rightarrow n^{\log_b a} \rightarrow n^{\log_2 4} \rightarrow n^2$

F(n) = n^2

$\rightarrow T(n) \in O(n^2 \log n)$

Example 3: $T(n) = 4T(n/2) + n^3 \Rightarrow T(n) \in ?$

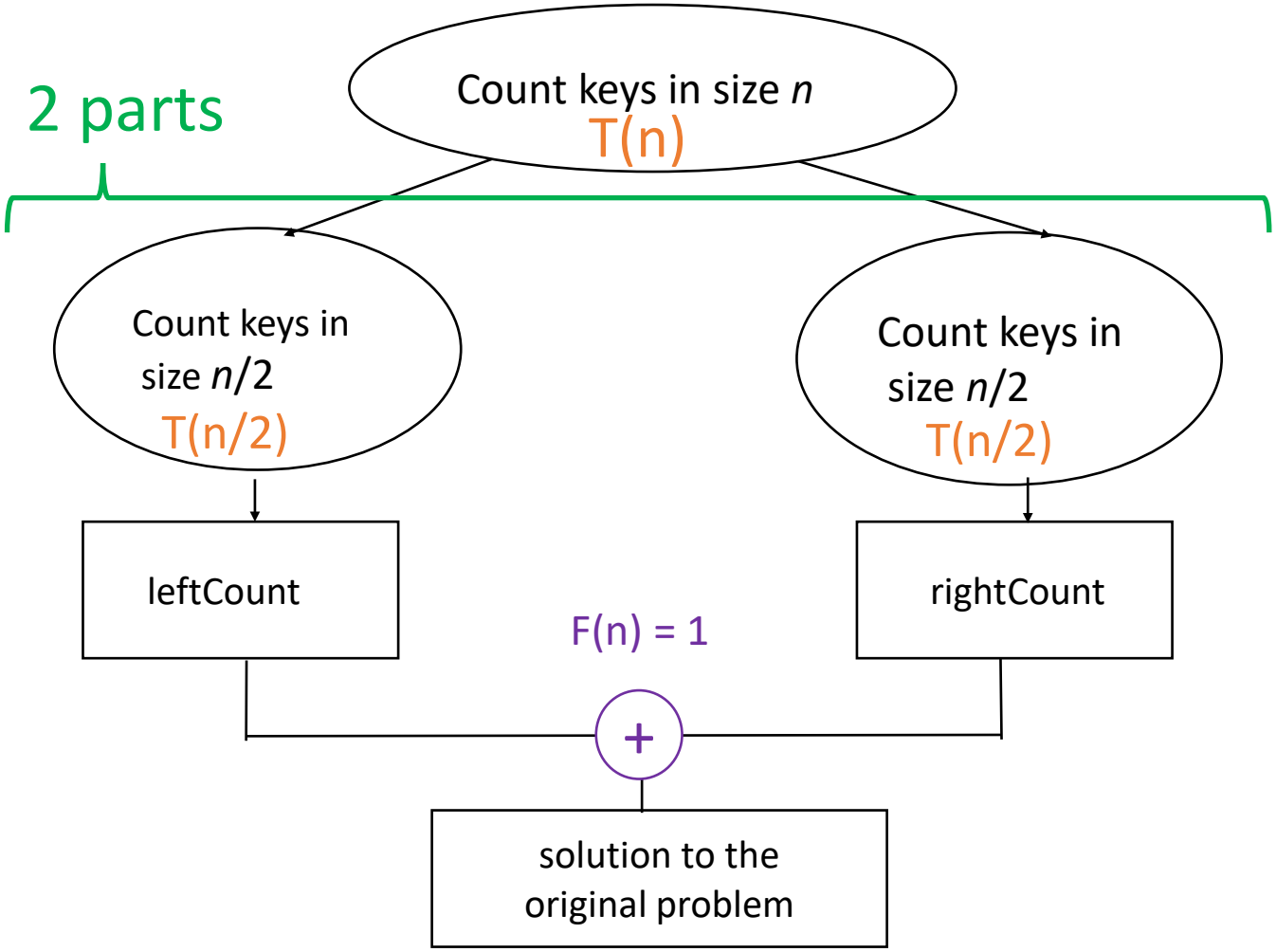
a = 4
b = 2
F(n) = n^3

$\rightarrow n^{\log_b a} \rightarrow n^{\log_2 4} \rightarrow n^2$

F(n) = n^3

$\rightarrow T(n) \in O(n^3)$

Back to CountKeys again



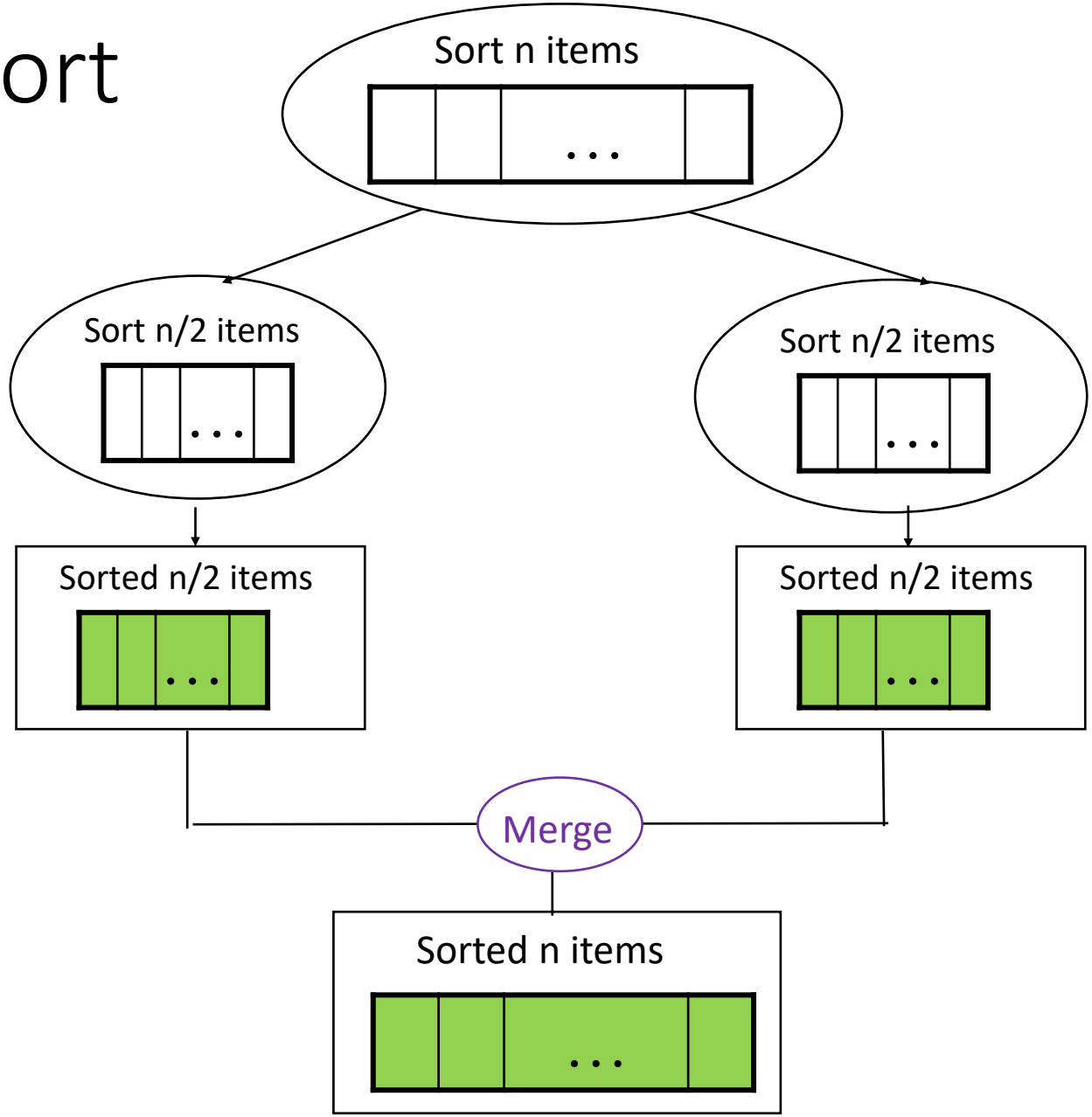
$$T(n) = 2\,T(n/2) + 1 \quad \Rightarrow \quad T(n) \in O(n)$$

Analyzing a *decrease and conquer* algorithm

- Can use the same Master Theorem
- Fake coin problem:
  - a=1 (we only solve ONE sub-part)
  - b=2 (each part is n/2)
  - F(n)=1 (no combination step = constant time)
- → Running time is $T(n) = 1T(n/2) + 1$
  - $n^{\log_b a} == n^{\log_2 1} == n^0 == 1$
  - $F(n) = 1$
  - They are equal
- → Final answer is $O(n^{\log_b a} \log n) == O(\log n)$

Mergesort

Mergesort


Sort n items
Sort n/2 items
Sort n/2 items
Sorted n/2 items
Sorted n/2 items
Merge
Sorted n items

Pseudocode of Mergesort

ALGORITHM *Mergesort*$(A[0..n-1])$

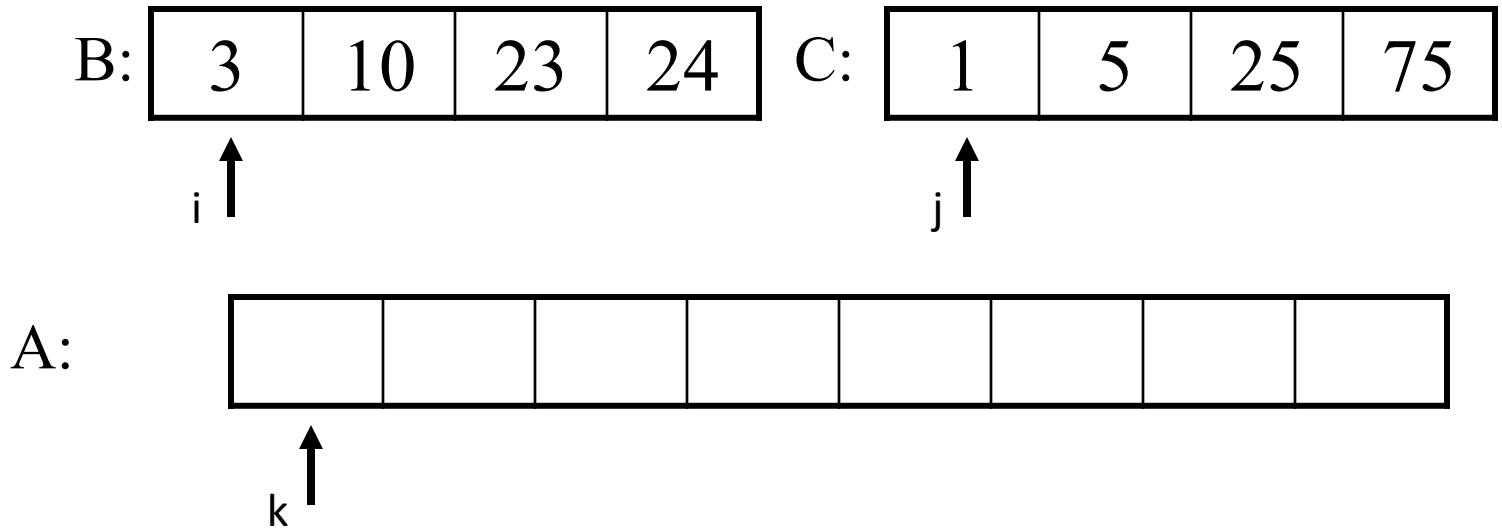
//Sorts array $A[0..n-1]$ by recursive mergesort
//Input: An array $A[0..n-1]$ of orderable elements
//Output: Array $A[0..n-1]$ sorted in nondecreasing order
if $n > 1$
    copy $A[0..\lfloor n/2 \rfloor - 1]$ to $B[0..\lfloor n/2 \rfloor - 1]$
    copy $A[\lfloor n/2 \rfloor..n-1]$ to $C[0..\lceil n/2 \rceil - 1]$
    Mergesort$(B[0..\lfloor n/2 \rfloor - 1])$
    Mergesort$(C[0..\lceil n/2 \rceil - 1])$
    Merge(B, C, A)

Mergesort

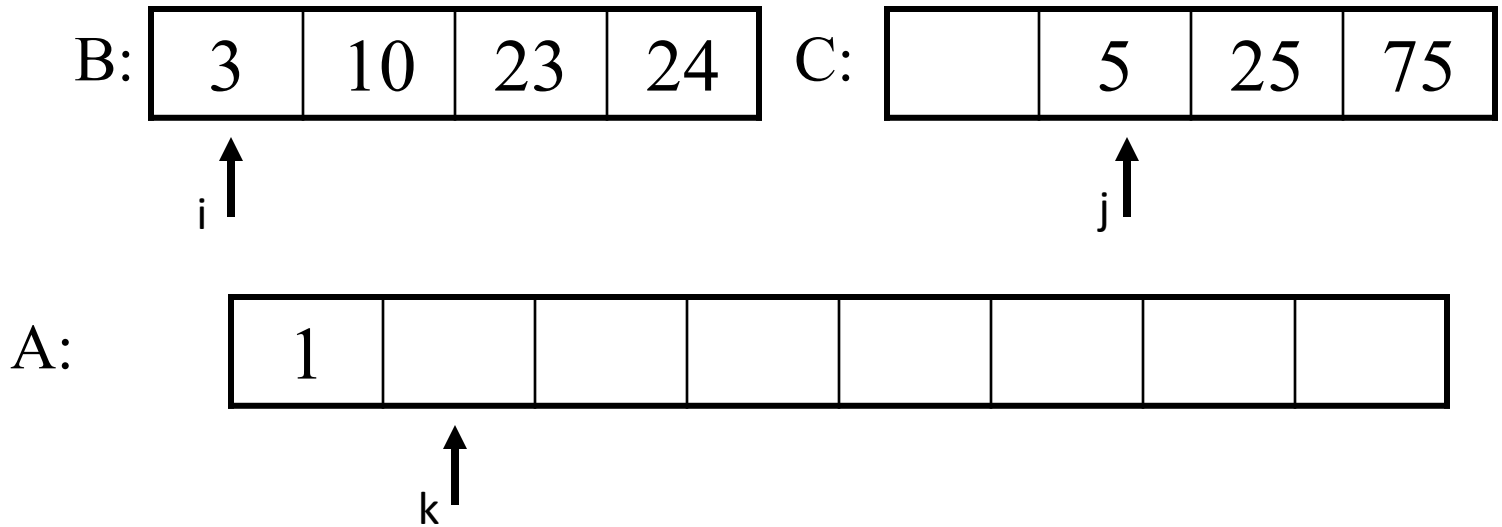
- The “combine partial solutions” part of mergesort is *merging two sorted arrays into one*

- Example:
  - B = { 3 8 9 } C = { 1 5 7 }
  - merge(B, C) = { 1 3 5 7 8 9 }

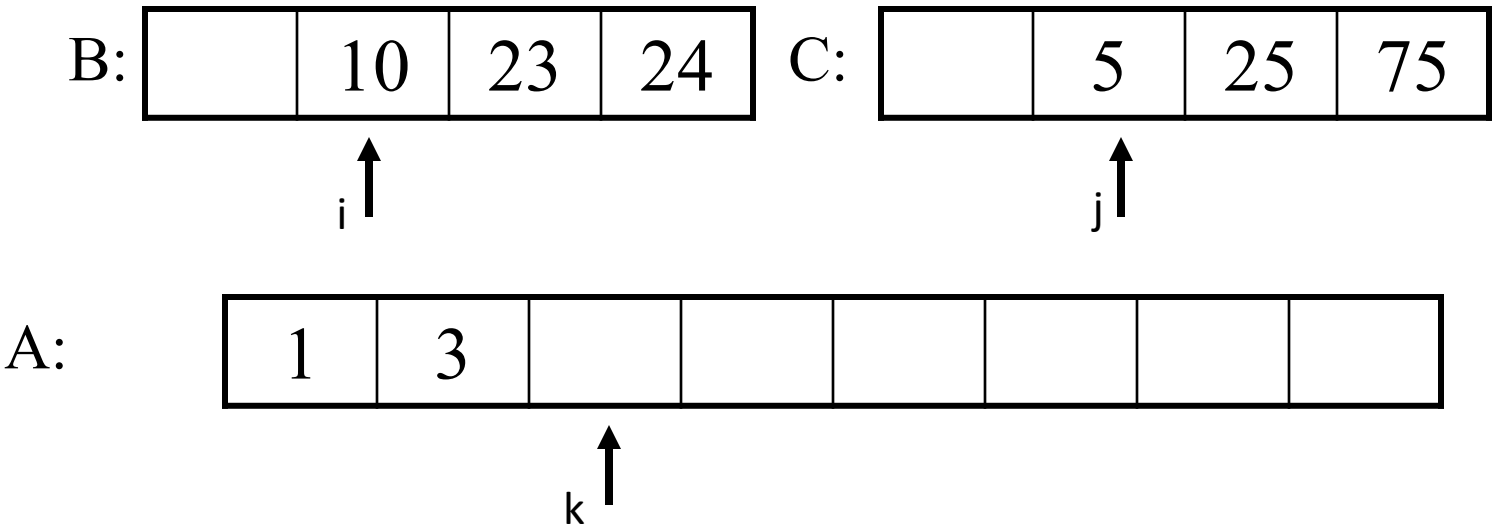
Merging



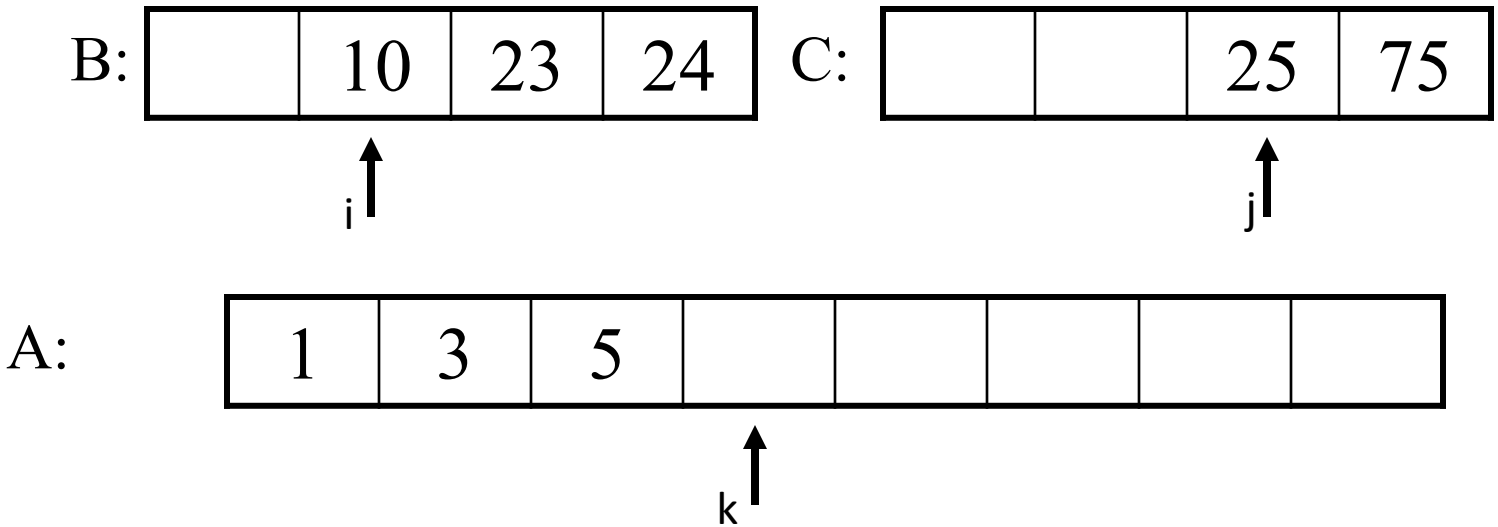
Merging (cont.)



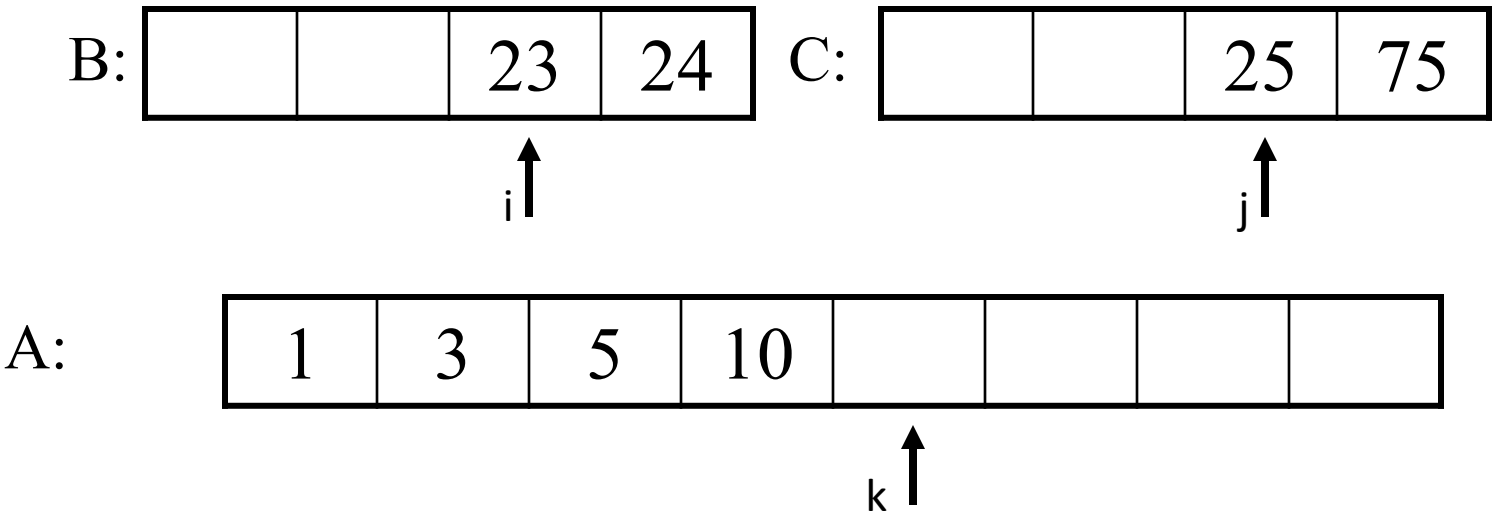
Merging (cont.)



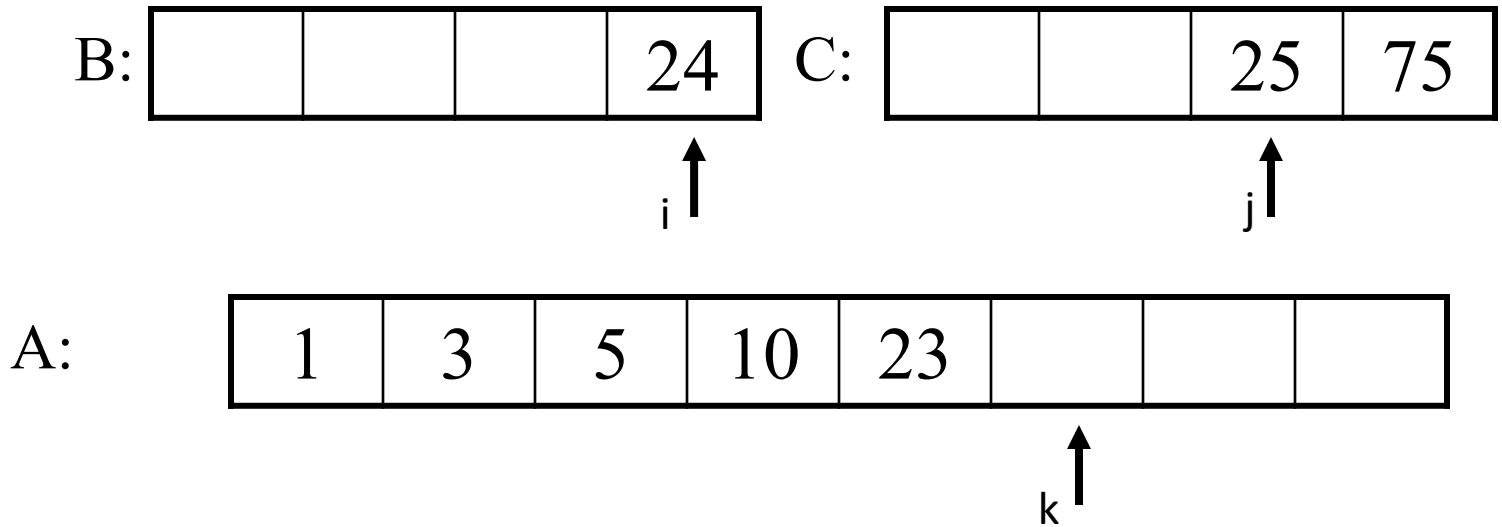
Merging (cont.)



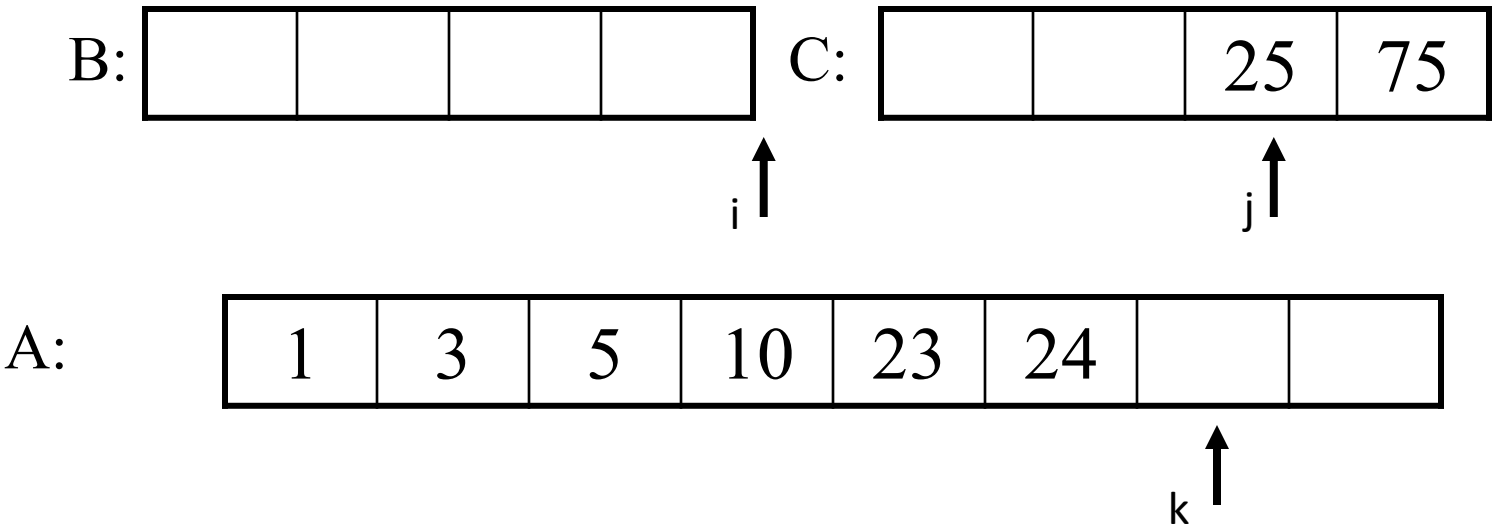
Merging (cont.)



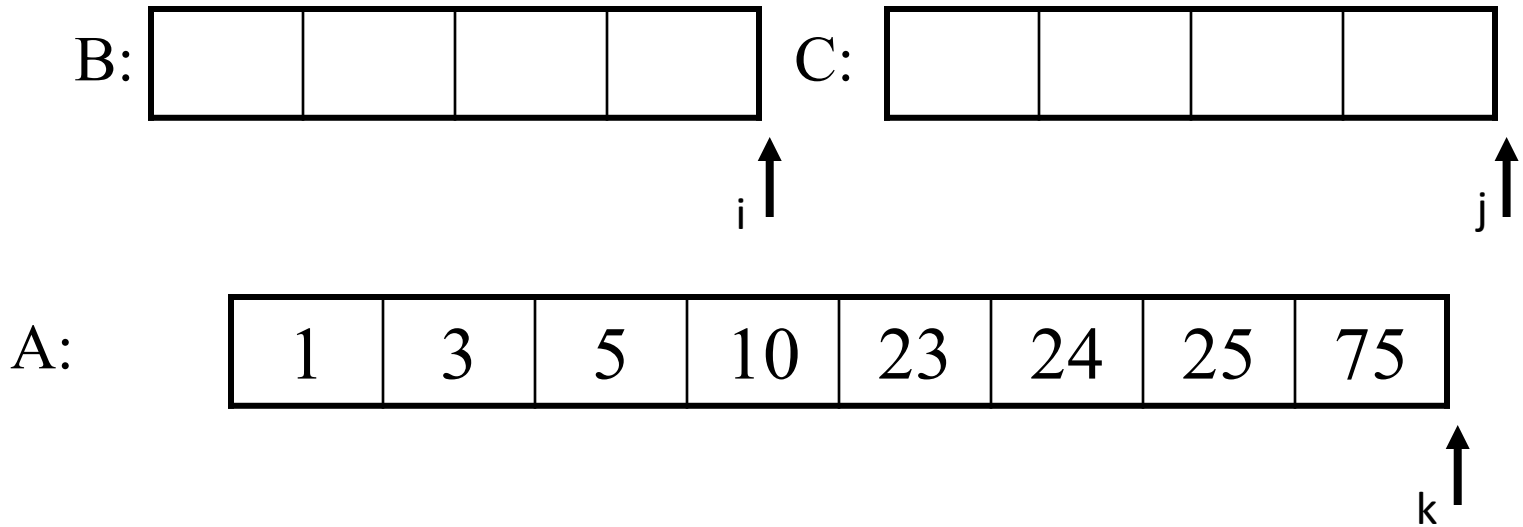
Merging (cont.)



Merging (cont.)



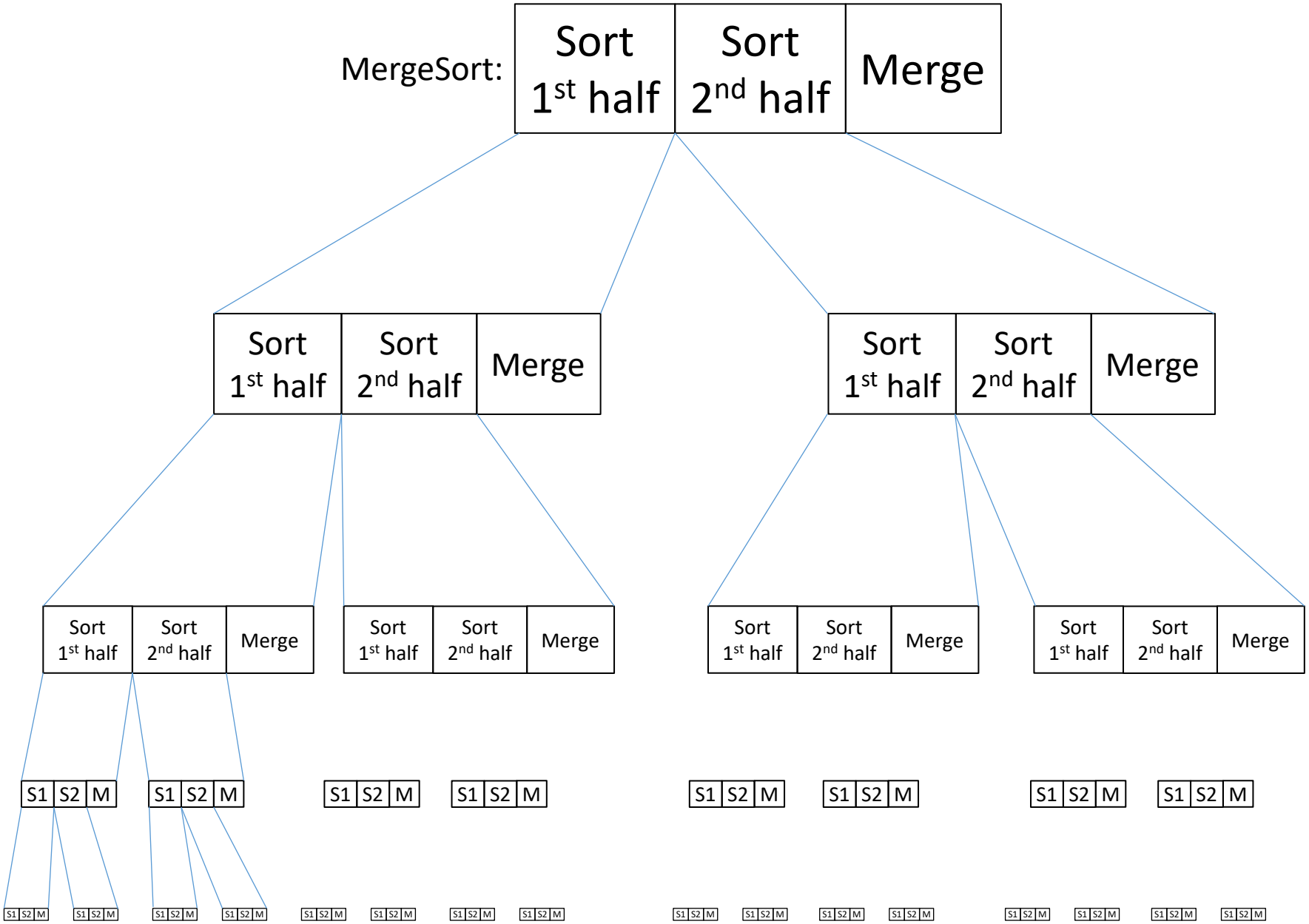
Merging (cont.)



Pseudocode of Merge

ALGORITHM $Merge(B[0..p-1], C[0..q-1], A[0..p+q-1])$

```
//Merges two sorted arrays into one sorted array
//Input: Arrays B[0..p − 1] and C[0..q − 1] both sorted
//Output: Sorted array A[0..p + q − 1] of the elements of B and C
i ← 0; j ← 0; k ← 0
while i < p and j < q do
    if B[i] ≤ C[j]
        A[k] ← B[i]; i ← i + 1
    else A[k] ← C[j]; j ← j + 1
    k ← k + 1
if i = p
    copy C[j..q − 1] to A[k..p + q − 1]
else copy B[i..p − 1] to A[k..p + q − 1]
```


MergeSort:
Sort 1st half
Sort 2nd half
Merge
Sort 1st half
Sort 2nd half
Merge
Sort 1st half
Sort 2nd half
Merge
Sort 1st half
Sort 2nd half
Merge
Sort 1st half
Sort 2nd half
Merge
Sort 1st half
Sort 2nd half
Merge
Sort 1st half
Sort 2nd half
Merge
S1 S2 M
S1 S2 M
S1 S2 M
S1 S2 M
S1 S2 M
S1 S2 M
S1 S2 M
S1 S2 M

Mergesort Example

99	6	86	15	58	35	86	4	0

99	6	86	15

58	35	86	4	0

99	6

86	15

58	35

86	4	0

99

6

86

15

58

35

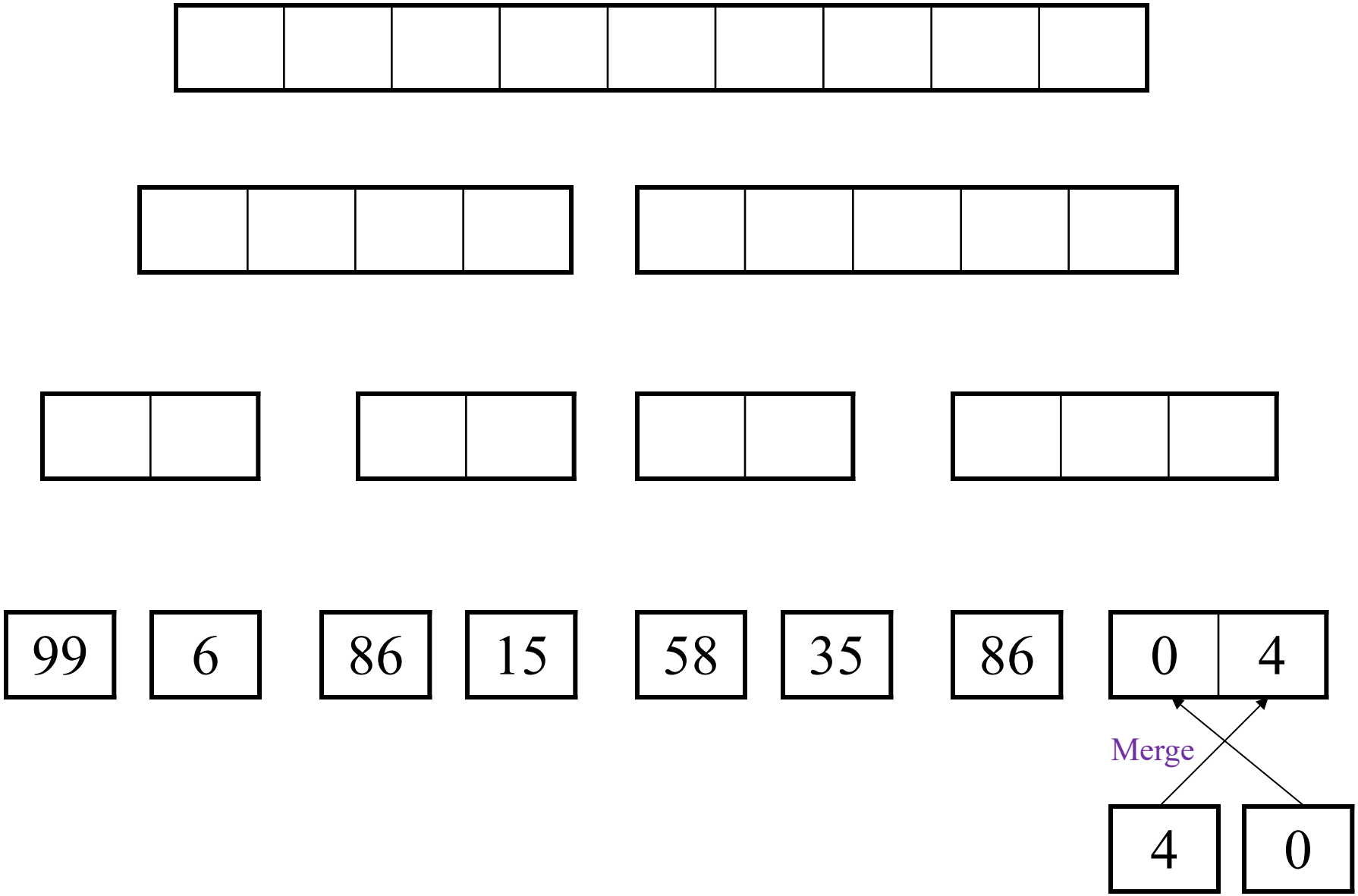
86

4	0

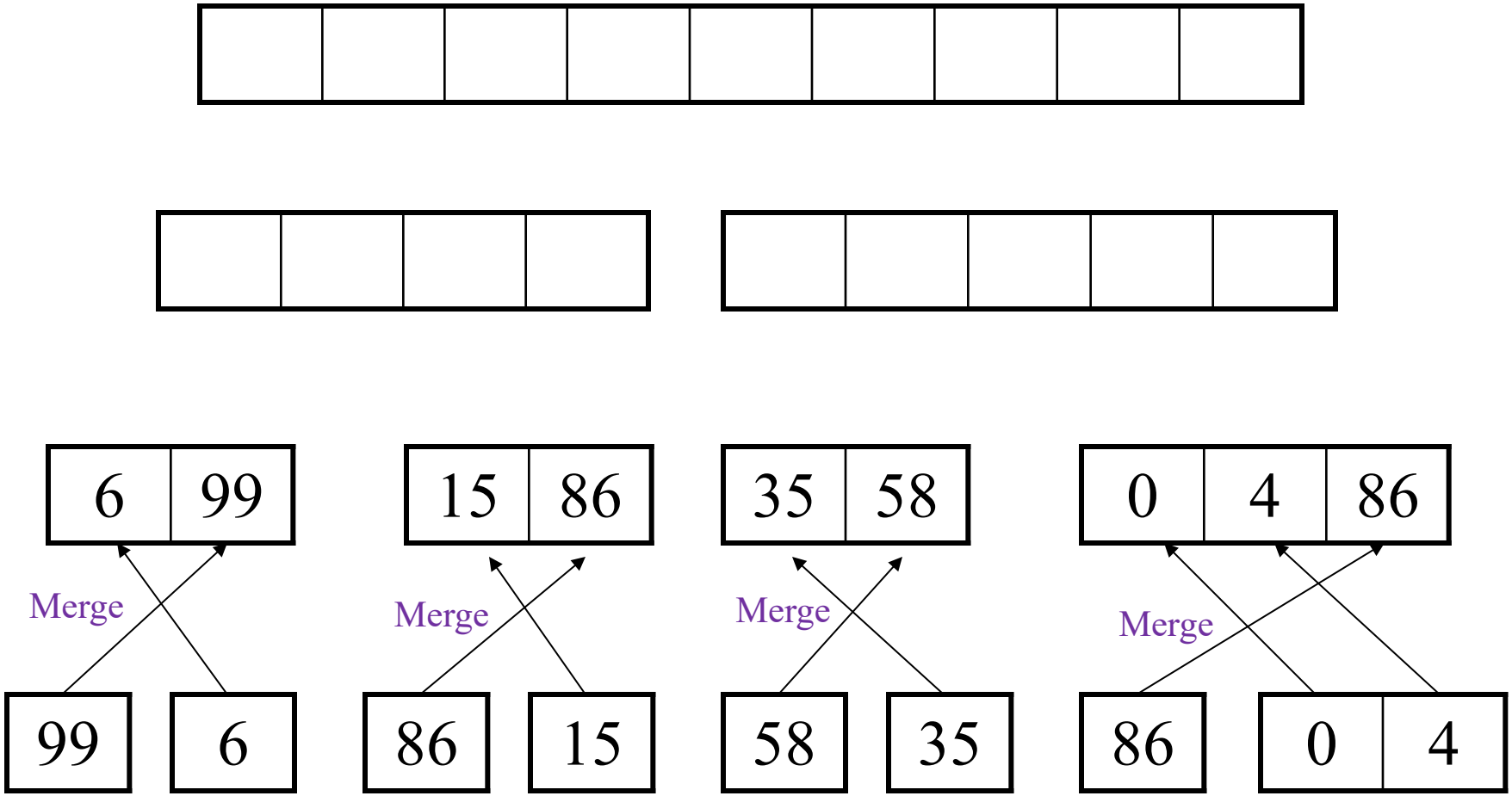
4

0

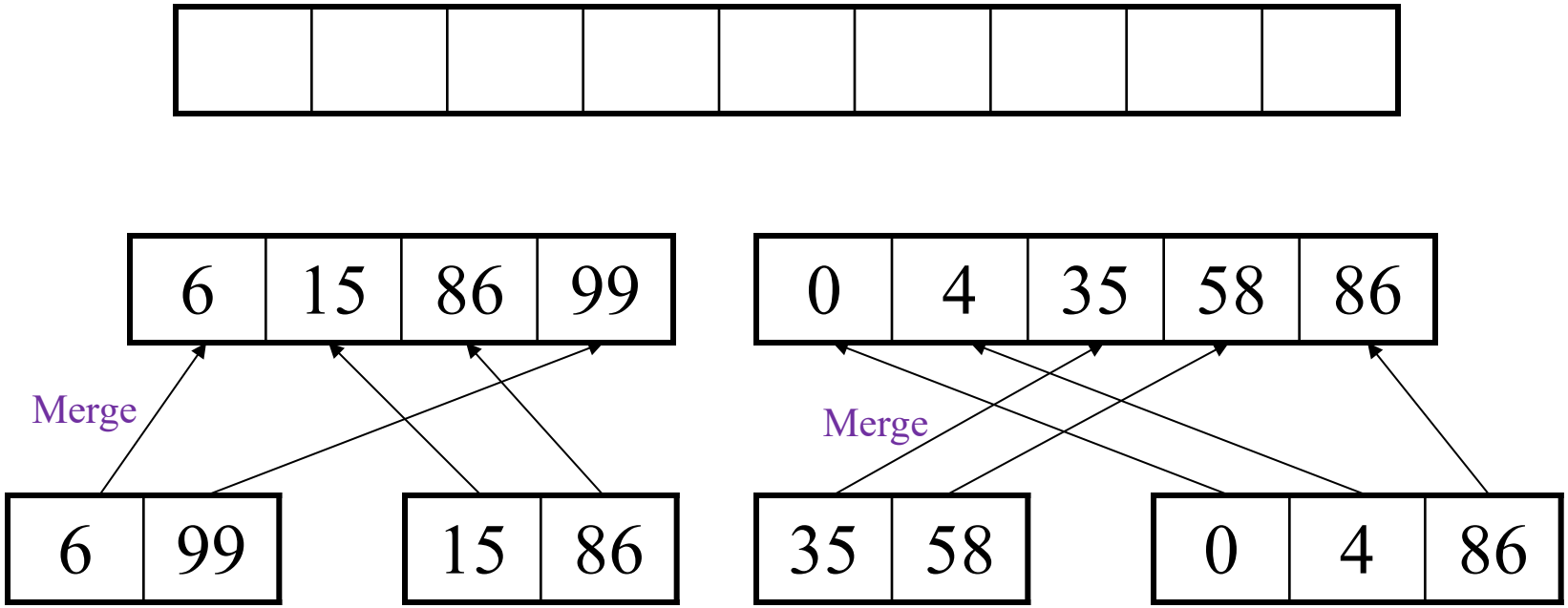
Mergesort Example



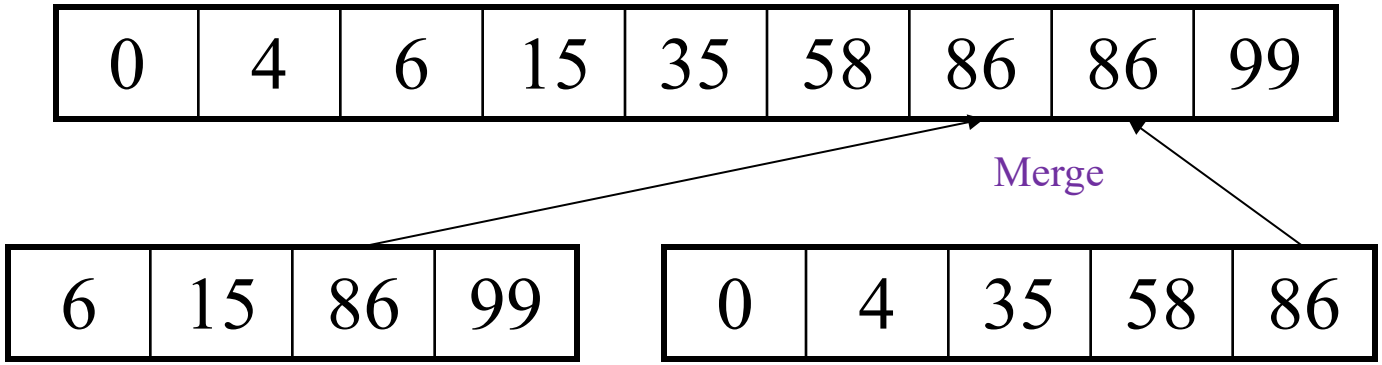
Mergesort Example



Mergesort Example



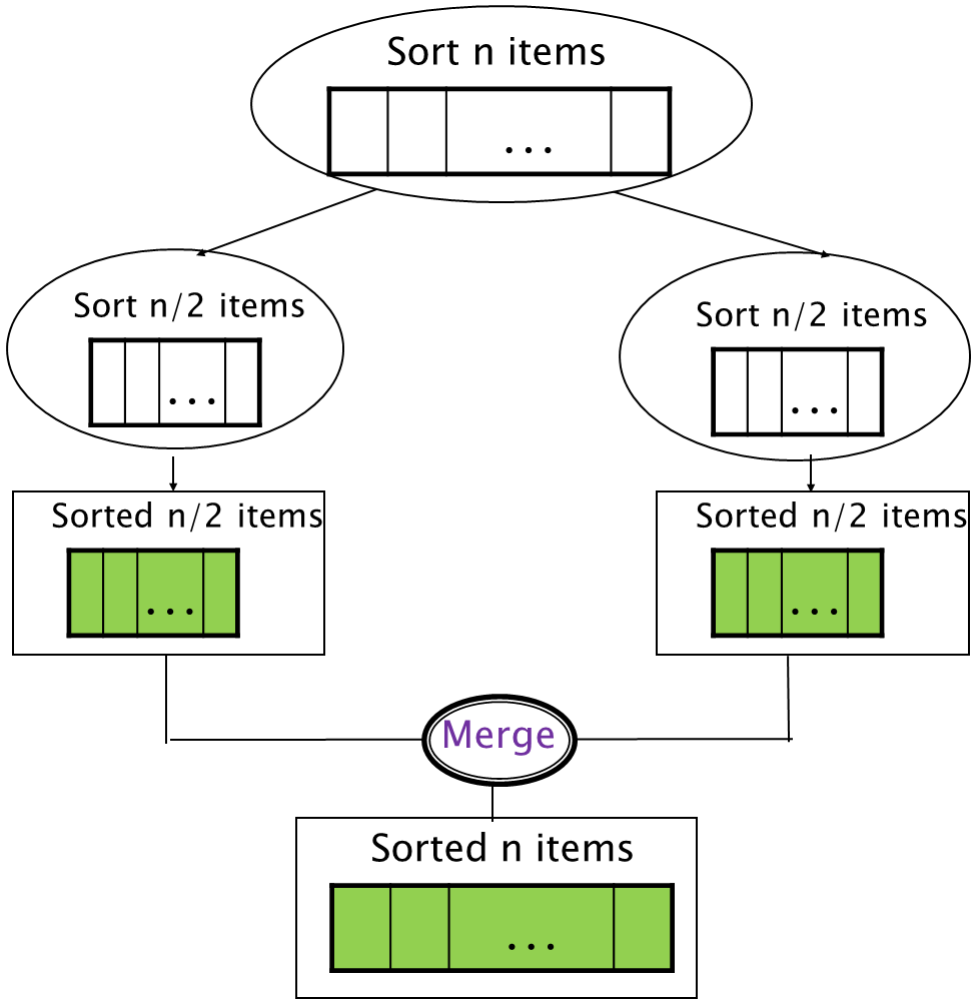
Mergesort Example



Mergesort Example

0	4	6	15	35	58	86	86	99

Mergesort running time



$T(n) = 2\,T(n/2) + n$

Compare the two key expressions:

$n^{\log_b a} = n^{\log_2 2} = n$

$F(n) = n$

They are equal!

$\rightarrow T(n) \in O(n \log n)$

Practice problems

1. Chapter 5.1, page 174, questions 1, 2, 6
2. Chapter 5.3, page 185, question 2
3. Implement a function to check if a tree is balanced. A balanced tree is defined to be a tree such that no two leaf nodes differ in distance from the root by more than one.